

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

admin

config

hospital

patients

local

>_MONGOSH

> use hospital

< switched to db hospital

> db.createCollection("Patient")

< { ok: 1 }

> db.patients.insertMany([

{ patientName: "Arun", doctor: "Dr. Ravi", department: "Cardiology", age: 45,

treatmentCost: 20000, daysAdmitted: 5 },

{ patientName: "Meena", doctor: "Dr. Kumar", department: "Neurology", age: 50,

treatmentCost: 30000, daysAdmitted: 7 },

{ patientName: "Raj", doctor: "Dr. Ravi", department: "Cardiology", age: 60,

treatmentCost: 15000, daysAdmitted: 3 },

{ patientName: "Divya", doctor: "Dr. Mehta", department: "Orthopedics", age: 35,

treatmentCost: 10000, daysAdmitted: 2 },

{ patientName: "Kiran", doctor: "Dr. Kumar", department: "Neurology", age: 40,

treatmentCost: 25000, daysAdmitted: 6 },

{ patientName: "Sneha", doctor: "Dr. Mehta", department: "Orthopedics", age: 30,

treatmentCost: 8000, daysAdmitted: 2 },

{ patientName: "Vijay", doctor: "Dr. Ravi", department: "Cardiology", age: 55,

treatmentCost: 22000, daysAdmitted: 4 },

{ patientName: "Priya", doctor: "Dr. Kumar", department: "Neurology", age: 48,

treatmentCost: 27000, daysAdmitted: 5 },

{ patientName: "Anil", doctor: "Dr. Mehta", department: "Orthopedics", age: 52,

treatmentCost: 12000, daysAdmitted: 3 },

>_MONGOSH

```
> db.patients.aggregate([
  {$match:{treatmentCost: {$gt: 20000}}},
  {$project:{ patientName: 1, department: 1, _id: 0 }}
])
< {
  patientName: 'Meena',
  department: 'Neurology'
}
{
  patientName: 'Kiran',
  department: 'Neurology'
}
{
  patientName: 'Vijay',
  department: 'Cardiology'
}
{
  patientName: 'Priya',
  department: 'Neurology'
}
```

>_MONGOSH

```
> db.patients.aggregate([
  {$sort:{treatmentCost: -1}}
])
< {
  _id: ObjectId('6a1090e1aae6be7df52cd538'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
  daysAdmitted: 7
}
{
  _id: ObjectId('6a1090e1aae6be7df52cd53e'),
  patientName: 'Priya',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 48,
  treatmentCost: 27000,
  daysAdmitted: 5
}
{
  _id: ObjectId('6a1090e1aae6be7df52cd53b'),
```

```
> db.patients.aggregate([
  {$limit: 5}
])
< {
  _id: ObjectId('6a1090e1aae6be7df52cd537'),
  patientName: 'Arun',
  doctor: 'Dr. Ravi',
  department: 'Cardiology',
  age: 45,
  treatmentCost: 20000,
  daysAdmitted: 5
}
{
  _id: ObjectId('6a1090e1aae6be7df52cd538'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
```

```
> db.patients.aggregate([
  {$skip: 3}
])
< {
  _id: ObjectId('6a1090e1aae6be7df52cd53a'),
  patientName: 'Divya',
  doctor: 'Dr. Mehta',
  department: 'Orthopedics',
  age: 35,
  treatmentCost: 10000,
  daysAdmitted: 2
}
{
  _id: ObjectId('6a1090e1aae6be7df52cd53b'),
  patientName: 'Kiran',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 40,
  treatmentCost: 25000,
  daysAdmitted: 6
}
{
  id: ObjectId('6a1090e1aae6be7df52cd53c'),
```

```
> db.patients.aggregate([
  {$match:{department: "Cardiology"}},
  {$project:{ patientName: 1, _id: 0 }}
])
< {
  patientName: 'Arun'
}
{
  patientName: 'Raj'
}
{
  patientName: 'Vijay'
}
{
  patientName: 'Riya'
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", totalTreatmentCost: {$sum: "$treatmentCost"}}}
])
< {
  _id: 'Orthopedics',
  totalTreatmentCost: 30000
}
{
  _id: 'Cardiology',
  totalTreatmentCost: 75000
}
{
  _id: 'Neurology',
  totalTreatmentCost: 82000
}
> db.patients.aggregate([
  {$group: {_id: "$doctor", averageTreatmentCost: {$avg: "$treatmentCost"}}},
  {$sort: {averageTreatmentCost: 1}}
])
< {
  _id: 'Dr. Mehta',
  averageTreatmentCost: 10000
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", totalTreatmentCost: {$sum: "$treatmentCost"}}},
  {$match: {totalTreatmentCost: {$gt: 50000}}}
])
```

```
< {
  _id: 'Neurology',
  totalTreatmentCost: 82000
}
```

```
{
  _id: 'Cardiology',
  totalTreatmentCost: 75000
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", patientCount: {$sum: 1}}},
  {$sort: {patientCount: -1}}
])
```

```
< {
  _id: 'Cardiology',
  patientCount: 4
}
```

```
{
  _id: 'Orthopedics',
  patientCount: 3
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$doctor", totalTreatmentCost: {$sum: "$treatmentCost"}}},
  {$sort: {totalTreatmentCost: -1}},
  {$limit: 3}
])
< {
  _id: 'Dr. Kumar',
  totalTreatmentCost: 82000
}
{
  _id: 'Dr. Ravi',
  totalTreatmentCost: 75000
}
{
  _id: 'Dr. Mehta',
  totalTreatmentCost: 30000
}
```

```
> db.patients.aggregate([
  {$match:{age:{$gt:40}}},
  {$group: {_id:"$department",patientCount:{$sum:1}}}
])
< {
  _id: 'Neurology',
  patientCount: 2
}
{
  _id: 'Cardiology',
  patientCount: 3
}
{
  _id: 'Orthopedics',
  patientCount: 1
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", averageTreatmentCost: {$avg: "$treatmentCost"}}},
  {$project: {department: "$_id", averageTreatmentCost: 1, _id: 0}},
  {$sort: {averageTreatmentCost: -1}}
])
< {
  averageTreatmentCost: 27333.333333333332,
  department: 'Neurology'
}
{
  averageTreatmentCost: 18750,
  department: 'Cardiology'
}
{
  averageTreatmentCost: 10000,
  department: 'Orthopedics'
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", patientCount: {$sum: 1}}},
  {$match: {patientCount: {$gt: 2}}}
])
< {
  _id: 'Orthopedics',
  patientCount: 3
}
{
  _id: 'Cardiology',
  patientCount: 4
}
{
  _id: 'Neurology',
  patientCount: 3
}
```

```
> db.patients.aggregate([
  {$group: {_id: "$department", totalTreatmentCost: {$sum: "$treatmentCost"}}},
  {$sort: {totalTreatmentCost: -1}},
  {$skip: 1},
  {$limit: 3}
])
< {
  _id: 'Cardiology',
  totalTreatmentCost: 75000
}
{
  _id: 'Orthopedics',
  totalTreatmentCost: 30000
}
> db.patients.aggregate([
  {$match: {treatmentCost: {$gt: 10000}}},
  {$group: {_id: "$doctor", totalTreatmentCost: {$sum: "$treatmentCost"}}},
  {$sort: {totalTreatmentCost: -1}},
  {$limit: 2}
])
< {
  _id: 'Dr. Kumar',
  totalTreatmentCost: 82000
}
```